

Pressure in Liquids and Air

Liquids and air push in all directions — and the deeper you go, the greater the pressure.

Liquids and air exert pressure

Pressure in liquids

- A liquid exerts pressure on the **bottom and the sides** of its container — in fact, in **all directions**.
- Liquid pressure **depends on the height (depth) of the column** — greater height, greater pressure. (It does *not* depend on the container's width.)

Everyday links: overhead tanks are placed high so taps get good pressure; a dam's base is broader because water pressure is largest near the bottom.

Pressure of air (atmospheric pressure)

The envelope of air around the Earth is the **atmosphere**. Air has weight and exerts **atmospheric pressure** on everything, in **all directions**.

- A **rubber sucker** sticks because outside air pressure is higher than the reduced pressure inside it.
- Atmospheric pressure is huge (the air column over a $15\text{ cm} \times 15\text{ cm}$ area pushes with about 2250 N), yet we are not crushed because the **pressure inside our body balances it**.

Practical units: air pressure is often given in **millibar (mb)** or **hectopascal (hPa)**; $1\text{ mb} = 1\text{ hPa} = 100\text{ Pa}$.

Where students lose marks

Trap 1 — Liquid pressure depends on height, not width. Two columns of the same height give the same pressure even if one is wider.

Trap 2 — Liquids/air press in ALL directions, not just downward.

Trap 3 — A sucker sticks due to outside air pressure, not "suction magic". The inside pressure is reduced.

Not crushed: our internal pressure equals atmospheric pressure, so it balances out.

Model answers — write them like this

Example A. Why are overhead water tanks placed high up?

Step 1: Liquid pressure increases with the height of the column.

Step 2: A taller column gives more pressure at the taps.

$\therefore$ Height increases pressure, giving a strong stream from the taps.

Example B. Why does a rubber sucker stick to a smooth wall?

Step 1: Pressing it pushes air out, lowering the pressure inside.

Step 2: Higher outside atmospheric pressure holds it against the wall.

$\therefore$ The pressure difference makes the sucker stick.

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